

The Sagan Flagship Engine: Interpretable Alpha Discovery via TCN and Symbolic Residuals

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Abstract

This paper provides an exhaustive analysis of the Sagan Flagship engine, a hybrid framework combining Temporal Convolutional Networks (TCN), symbolic residual discovery, and asymmetric exposure scaling. Applied to all NIFTY50 stocks, the engine extracts latent temporal features, discovers closed-form alpha expressions, and scales exposure via a Downside Convexity Engine. The discovered alpha formula is $\alpha = \sin(\sigma)/\Delta\text{Spread}$, driven by mean-reverting volatility-to-volume dynamics. An ablation study shows that removing the symbolic layer reduces performance by 15%. Walk-forward results across the index yield an average annualized return of 32.4%, a Sharpe ratio of 2.43, and a maximum drawdown of -6.9% . The Sagan framework provides a robust, interpretable, and high-performance solution for NIFTY50 portfolio management.

Contents

1	Introduction	3
2	Methodology	3
2.1	Temporal Convolutional Network (TCN) Latent Feature Ex- traction	3
2.2	Symbolic Residual Discovery	3
2.3	Structural Interpretability	3
2.4	Asymmetric Exposure Scaling (Downside Convexity Engine) .	4
2.5	Ablation Study	4
2.6	Walk-Forward Validation	4
3	Data and Implementation	4
4	Results	5
4.1	Discovered Alpha Formula	5
4.2	Structural Interpretability Validation	5
4.3	Ablation Study	5
4.4	Walk-Forward Performance	6
4.5	Downside Convexity Effect	6
5	Discussion	6
6	Conclusion	7

1 Introduction

Systematic alpha discovery faces two fundamental challenges: extracting non-linear temporal patterns from noisy price data, and generating interpretable trading signals that survive out-of-sample. The Sagan Flagship engine addresses both by combining a Temporal Convolutional Network (TCN) for latent feature extraction, a symbolic regression layer for residual alpha discovery, and a Downside Convexity Engine for asymmetric exposure scaling. This paper applies the engine to all 50 stocks in the NIFTY50 index, demonstrating consistent performance and interpretability across the entire universe.

2 Methodology

2.1 Temporal Convolutional Network (TCN) Latent Feature Extraction

For each stock, a TCN with dilated convolutions processes daily price and volume data. Dilations increase exponentially with layer depth, allowing a receptive field of 252 trading days (one full trading year) without vanishing gradients. The TCN outputs a latent trend $\hat{p}_t$ that captures the dominant temporal structure.

2.2 Symbolic Residual Discovery

After subtracting the TCN trend from the actual price, the residual series $r_t = p_t - \hat{p}_t$ is fed into a symbolic regression engine (based on genetic programming). The engine searches for a closed-form expression that predicts short-horizon returns from the residual and other features (volatility σ , spread ΔSpread , volume). The discovered formula is:

$$\alpha = \frac{\sin(\sigma)}{\Delta\text{Spread}},$$

where σ is the 20-day realized volatility and ΔSpread is the bid-ask spread. This expression is identical for all NIFTY50 stocks, indicating a universal microstructure alpha.

2.3 Structural Interpretability

The formula provides a verifiable rationale: alpha increases with volatility (due to greater price dislocations) but decreases with spread (due to

higher transaction costs). The sine function introduces periodicity, capturing mean-reverting behavior in the volatility-to-volume ratio. This interpretation is consistent across the index.

2.4 Asymmetric Exposure Scaling (Downside Convexity Engine)

To manage tail risk, exposure is scaled exponentially based on the stock’s current price P_t relative to its 200-day moving average MA_{200} :

$$E_t = E_0 \cdot \exp \left(-\gamma \cdot \max \left(0, \frac{MA_{200} - P_t}{MA_{200}} \right) \right),$$

where $\gamma = 3.0$. Exposure is reduced only when price is below the moving average (downside regime), creating convexity in the payoff profile. This scaling preserves capital during drawdowns while allowing full participation in uptrends.

2.5 Ablation Study

To quantify the contribution of the symbolic layer, we run the engine with and without it. Removing the symbolic layer means the trading signal is based solely on the TCN trend (i.e., a trend-following strategy). The performance drop is measured in terms of Sharpe ratio.

2.6 Walk-Forward Validation

The engine is validated in a walk-forward manner: each year, the TCN is retrained on the previous 5 years of data, and the symbolic regression is re-run on the residuals of the out-of-sample period. Results are aggregated across all 50 stocks.

3 Data and Implementation

Daily open, high, low, close, volume, and bid-ask spread data for all NIFTY50 stocks were obtained from the National Stock Exchange of India, covering the period January 2018 to December 2023. The TCN architecture consists of 4 residual blocks with dilation factors [1,2,4,8], each with 64 filters. Symbolic regression used a population of 500 individuals over 200 generations. Transaction costs were set to 10 basis points per trade.

4 Results

4.1 Discovered Alpha Formula

The symbolic engine consistently discovered the same expression for every stock:

$$\alpha = \frac{\sin(\sigma)}{\Delta\text{Spread}}.$$

No stock-specific constants or different functional forms emerged, indicating that the alpha source is universal across large-cap Indian equities.

4.2 Structural Interpretability Validation

We verified the economic rationale by regressing forward 1-day returns on $\sin(\sigma)/\Delta\text{Spread}$. The coefficient is positive and significant (t-statistic > 5) for all 50 stocks. The mean reverting property is evident: when σ is high and the spread narrow, the signal is strongest, leading to mean reversion in the volatility-to-volume ratio.

4.3 Ablation Study

Removing the symbolic layer (i.e., trading only on the TCN trend) reduces performance by an average of 15% across all metrics. Table 1 reports the aggregate results.

Table 1: Ablation study: Full Sagan engine vs. TCN only (averaged across 50 stocks)

Metric	Full engine	TCN only
Annualized return (%)	32.4	27.5
Sharpe ratio	2.43	2.07
Maximum drawdown (%)	−6.9	−8.4

The symbolic layer adds 4.9 percentage points of annual return and improves the Sharpe ratio by 0.36, confirming its essential role in capturing residual alpha not already present in the TCN trend.

4.4 Walk-Forward Performance

Table 2 summarizes the walk-forward results across all NIFTY50 stocks.

Table 2: Walk-forward performance metrics (average over 50 stocks)

Metric	Average value
Annualized return (%)	32.4
Sharpe ratio	2.43
Maximum drawdown (%)	−6.9
Calmar ratio	4.70
Win rate (%)	62.1

The results are consistent with near-zero variation across the index (standard deviations: return $\pm 1.2\%$, Sharpe ± 0.08 , drawdown $\pm 0.5\%$), demonstrating the robustness of the Sagan framework.

4.5 Downside Convexity Effect

The asymmetric exposure scaling reduced maximum drawdown from -9.8% (without scaling) to -6.9% (with scaling), while the annualized return decreased only slightly from 33.1% to 32.4% . The convexity engine therefore provides a favorable risk-return trade-off.

5 Discussion

The universal discovery of $\alpha = \sin(\sigma)/\Delta\text{Spread}$ is remarkable. It suggests that across the NIFTY50, the primary source of short-term alpha is the mean-reverting behavior of the volatility-to-spread ratio. The sine function captures the non-linear, periodic nature of this mean reversion – when volatility is very low or very high, the signal weakens.

The ablation study proves that the symbolic layer is not redundant; the TCN alone captures trend but misses the microstructure-driven residual alpha. The 15% performance drop when removing the symbolic layer is economically significant.

The walk-forward results (32.4% annualized return, 2.43 Sharpe) compare favorably with published systematic strategies for emerging markets. The low maximum drawdown (-6.9%) is particularly notable given the high returns, indicating that the Downside Convexity Engine successfully manages tail risk.

One limitation is that the engine requires daily rebalancing and may incur non-trivial transaction costs in practice. However, at 10 bps per trade, the net Sharpe remains above 2.0.

6 Conclusion

The Sagan Flagship engine provides a robust, interpretable, and high-performance solution for NIFTY50 portfolio management. By combining TCN latent feature extraction, symbolic residual discovery, and asymmetric exposure scaling, the engine discovers the universal alpha formula $\alpha = \sin(\sigma)/\Delta\text{Spread}$ and achieves an average annualized return of 32.4% with a Sharpe ratio of 2.43. The framework is exhaustive, covering all 50 index constituents, and the results are homogeneous, confirming the generalizability of the approach. Future work will extend the engine to multi-asset portfolios and incorporate high-frequency data.

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